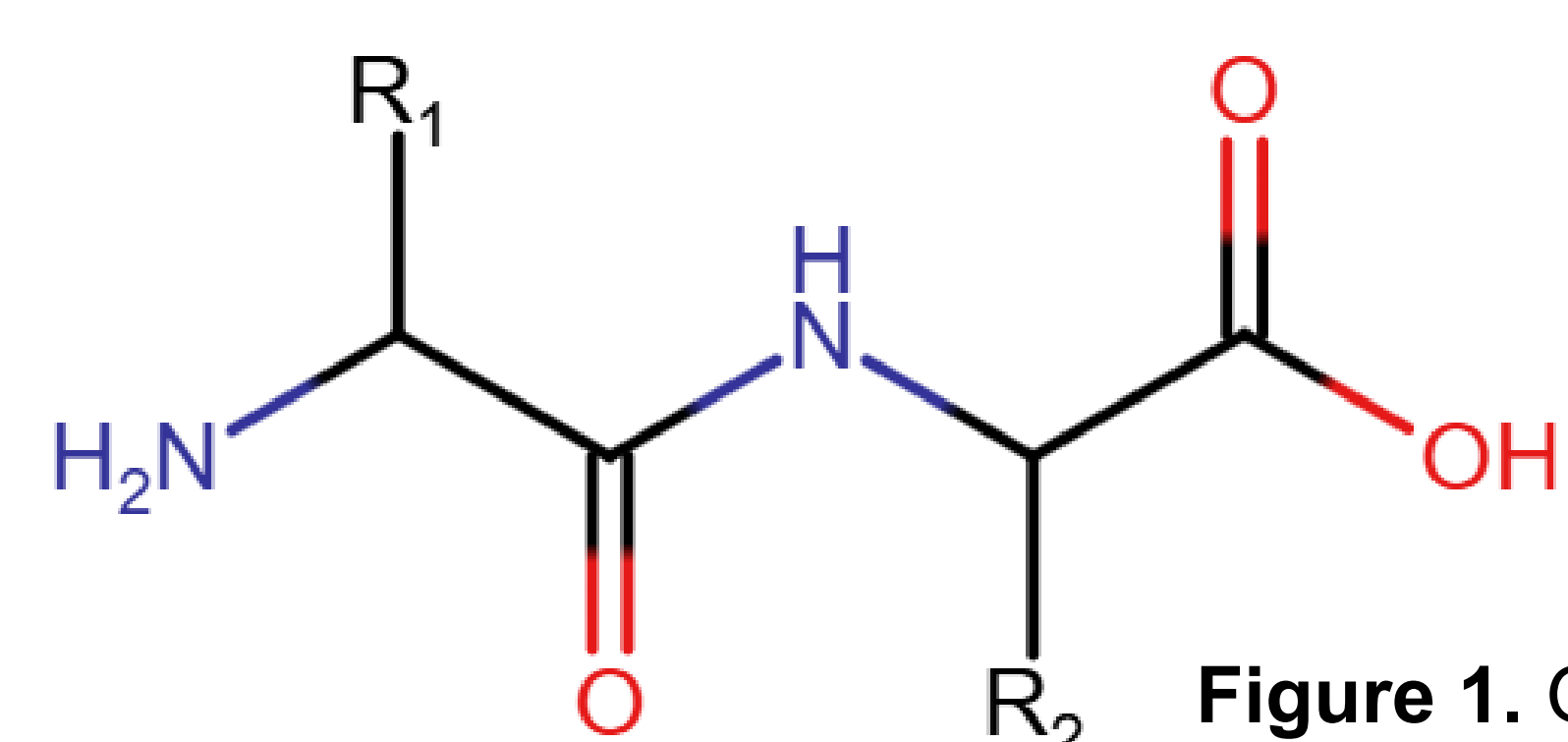


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Calculating Optimized Reference Structures for Algorithmic Determination of Protein Primary Structure

Background and Introduction

• Amino acids (AAs) are the monomeric building blocks of peptides and proteins, which are relatively very large



Peptide Structure:

- Amide/peptide bond connects amino acids
- R = amino acid side chains (e.g. $-\text{CH}_3$, $-\text{Ph}$, etc.)
- Each atom has a unique xyz (Cartesian) coordinate in space
- N-terminus: start of a peptide/protein
- C-terminus: end of a peptide/protein

Figure 1. Generic dipeptide structure, where R_1 and R_2 represent different side chains.

• Studying proteins experimentally (3+ options) is expensive (materials and waste disposal), time-consuming, and can be error-prone

1. Edman degradation (1950) [1]
 - a. Limited to about 50-70 AAs
 - b. Process of labeling and cleaving AAs from N- to C-term
 - c. N-term must be unmodified to start
 2. Tandem mass spectroscopy (MS/MS) [2]
 - a. Produces protein fragments
 - b. Lots of preparatory steps
 - c. Can't control where peptides cleave, thus requiring some trial and error
 3. Deriving protein sequence from (c/database-available)DNA or mRNA sequence [3]
 - a. *Easier than working with mRNA (cDNA = more stable)*
 - b. *Technical process, lots of steps = room for error*
 4. Studying proteins computationally is relatively cheaper, faster, and far less error-prone
- Important to know exact protein structure because structure = function

• Most programs that determine protein structure require primary structure to already be figured out. However, most *ab initio* chemistry papers only report xyz (Cartesian) coordinates of the atoms

Goal and Significance

Goal: Calculate small, realistic, optimized reference structures (amino acid monomers, homodimers, and homotrimers) for use in testing our algorithm that determines primary structure or AA sequence.

We seek to find the optimized geometry for each structure because that's often the lowest energy structure, and therefore the most stable structure, which is what nature prefers.

Significance: Since most *ab initio* chemistry papers only report xyz coordinates, but most programs require primary structure as an input, this testing data is useful in validating our program, which aims to convert xyz coordinates to the primary structure.

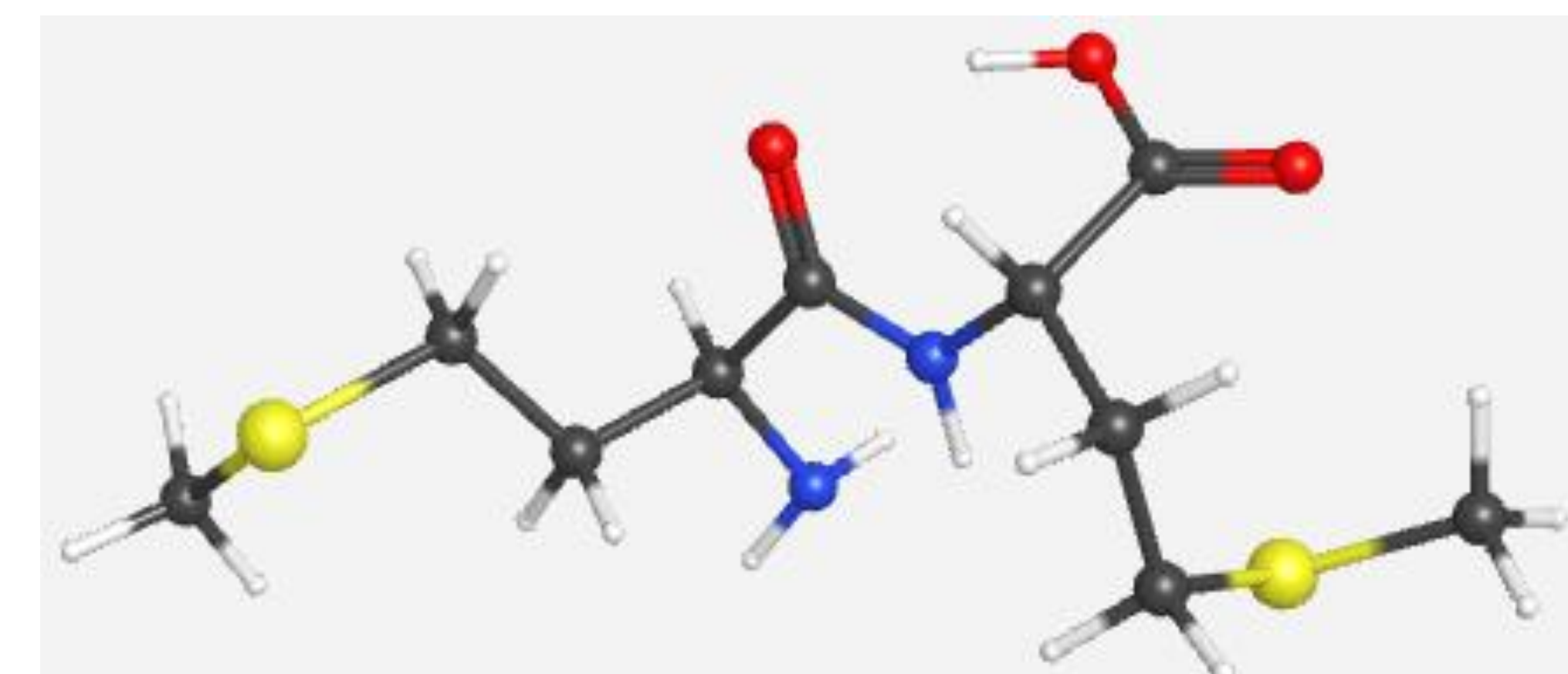
References

- Marvin was used for drawing, displaying, and characterizing chemical structures, substructures, and reactions, Marvin 22.11.1, **2022**, ChemAxon (<http://www.chemaxon.com>)
- [1] Edman, P.; Högfeldt, E.; Sillén, L. G.; Kinell, P.-O. Method for Determination of the Amino Acid Sequence in Peptides. *Acta Chem. Scand.* **1950**, 4, 283–293. <https://doi.org/10.3891/acta.chem.scand.04-0283>.
- [2] Klein, J. B.; A. Knepper, M. Protein Mass Spectrometry Made Simple. *J. Am. Soc. Nephrol. JASN* **2018**, 29 (6), 1585–1587. <https://doi.org/10.1681/ASN.2018030244>.
- [3] cDNA Production. <https://www.bio.davidson.edu/courses/genomics/method/cdnaproduction.html> (accessed 2026-07-20).

Methods

1. Create molecule in WebMO
 - a. Uses a built-in rudimentary geometry optimization by way of a quick DFT calculation/optimization

Figure 2. Molecular structure of dimethionine (diM) visualized and quickly optimized using WebMO. Black = carbon, white = hydrogen, red = oxygen, blue = nitrogen, yellow = sulfur.



2. Send WebMO xyz coordinates through a parser
 - a. Cleans up coordinates to a format that NWChem can understand (standard xyz format with columns: atom | x | y | z)

Figure 3. Screenshot of parser mentioned above. This was used to clean up output from WebMO (Figure 4) into a more standard output, seen in Figure 5.

```
1  with open(  
2  ) as f:  
3  content = f.read()  
4  cleaned_content = (  
5  content.replace(" ", "  
6  .replace("\\n", "  
7  .replace("\\", "  
8  .replace(", ", "  
9  .replace(" ", "  
10 )  
11 # noqa  
12  
13 print(cleaned_content)  
14  
15
```

Figure 4. Screenshot of WebMO xyz output for diM.

Results, Conclusion, Future Work, and Takeaways

Results: Out of 60 calculations, all but 5 converged (i.e. they did not find a lowest energy; those were the homotrimers of Cys, Phe, Gly, Asn, Gln). Output from NWChem looked like Figure 6.

Conclusion: There is no trend among non-converged AAs – further investigation is needed. No conclusions can be made.

Future work: Recalculate non-converged AAs. If again they do not converge, comb through full NWChem output and identify issues or trends, and address those.

Takeaways: Python, NWChem, and Linux skills; value of collaboration with lab-mates and those working on other projects.

Figure 5. Screenshot of output from parser for diM.

```
1  C,0.1231095997,-0.5641416989,0.0644368834  
2  N,0.1530221232,-1.1984128979,1.373755108  
3  H,1.089012322,-1.4200060682,1.6500500257  
4  H,-0.233907019,-0.6023459367,2.0805723408  
5  C,-1.2970577849,-0.5846931046,-0.4933063324  
6  N,-0.0590364355,-1.5591572657,-0.0417061804  
7  C,-3.4358079113,-1.8003953033,-0.5509744143  
8  C,-4.4724618253,-0.827765799,0.0341527922  
9  O,-4.1529818114,0.450607018,-0.0520477302  
10 H,-3.3564192998,0.5978268423,-0.5630686152  
11 O,-5.184811219,-1.1714112602,0.4736169295  
12 H,-3.4107694245,-1.5585765286,-1.6083374545  
13 C,-3.8472940729,-3.2776860125,-0.4267950679  
14 C,-3.3439414262,-4.0825595843,0.7803277882  
15 S,-3.6518358575,-3.4426161406,2.4528935656  
16 C,-5.4411665015,-3.660670371,2.6299021666  
17 H,-5.7125597035,-4.6976440594,2.4651961503  
18 H,-5.6922809834,-3.386317312,2.6372172101  
19 H,-5.9761458630,-3.0136045529,1.0204504702  
20 H,-3.7624263807,-5.0801360486,0.7129707616  
21 H,-2.2685498862,-4.2080499202,0.7262281685  
22 H,-4.9262799195,-3.3175514907,-0.4724295314  
23 H,-3.4754189901,-3.7963296669,-1.3059365617  
24 H,-1.7610155798,-2.0758902654,0.7379783904  
25 O,-1.656084644,0.2342985099,-1.3146617215  
26 H,0.4082335877,0.4844043093,0.089958784  
27 C,1.074848914,-1.3177432188,-0.8838985611  
28 C,1.2381165183,-0.6684827047,-2.2589171549  
29 S,2.365859917,-1.5619196988,-3.3649017993  
30 C,3.9821675344,-1.144681915,-2.6583250753  
31 H,4.7721250442,-1.5686711051,-3.313216541  
32 H,4.1232362689,-0.0715710481,-2.6213750674  
33 H,4.1165583159,-1.5638047843,-1.6697979648  
34 H,0.2036964087,-0.634409222,-2.782317345  
35 H,1.5772648473,0.3566689529,-2.1629566628  
36 H,2.0425730053,-1.3645818421,-0.3925910447  
37 H,0.7314601077,-2.3486429199,-0.9936951472
```

Figure 6. Screenshot of optimized xyz coordinate output from NWChem for diM.